

We built a scanning tunneling microscope in a basement

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Abstract

We detected tunneling, but weren't able to maintain tunneling range for long enough to get an image.

Every subsystem a tunneling microscope needs is built, working and calibrated. The amplifier resolves picoamps, the chain agrees with theory to **0.13 of a standard error**, and a real junction responds to voltage and to height.

Self-directed. No faculty advisor, no grant, no laboratory. **Eleven weeks of planning and design, eight weeks of building.** First power-on 29 August 2026; last bench session 19 September 2026. 29 session logs, 163 raw data files, 92 photographs of our own hardware and every analysis script are public — including the results we later withdrew.

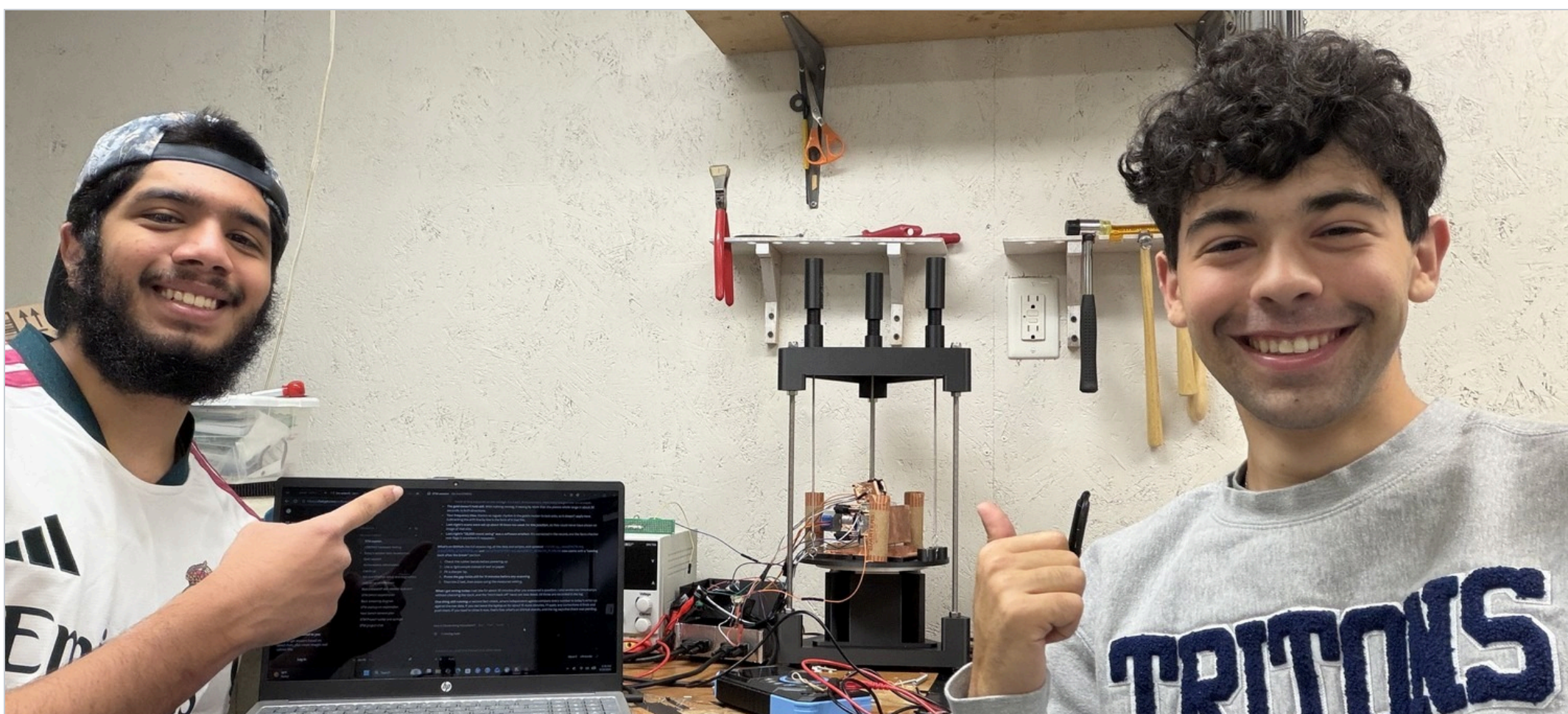
Introduction

Two undergraduates. A basement. Our first soldering.

Neither of us writes code, and neither had much electronics experience — **this project was the first real soldering either of us has done.** It was built and run in the basement of a house people live in, with people moving around and no control over the air or the temperature. A wooden bench on a concrete floor, a few meters from the breaker panel. **Not a lab. Not a clean room.**

This is not a disclaimer. It is the result. A tunneling microscope is normally a clean-room instrument — and the blocker we found and measured is exactly the class of problem those rooms exist to remove. Stated up front, our limitation becomes a measurement of what this environment costs you.

Conditions stated by Jacob Katz, 2026-09-19.

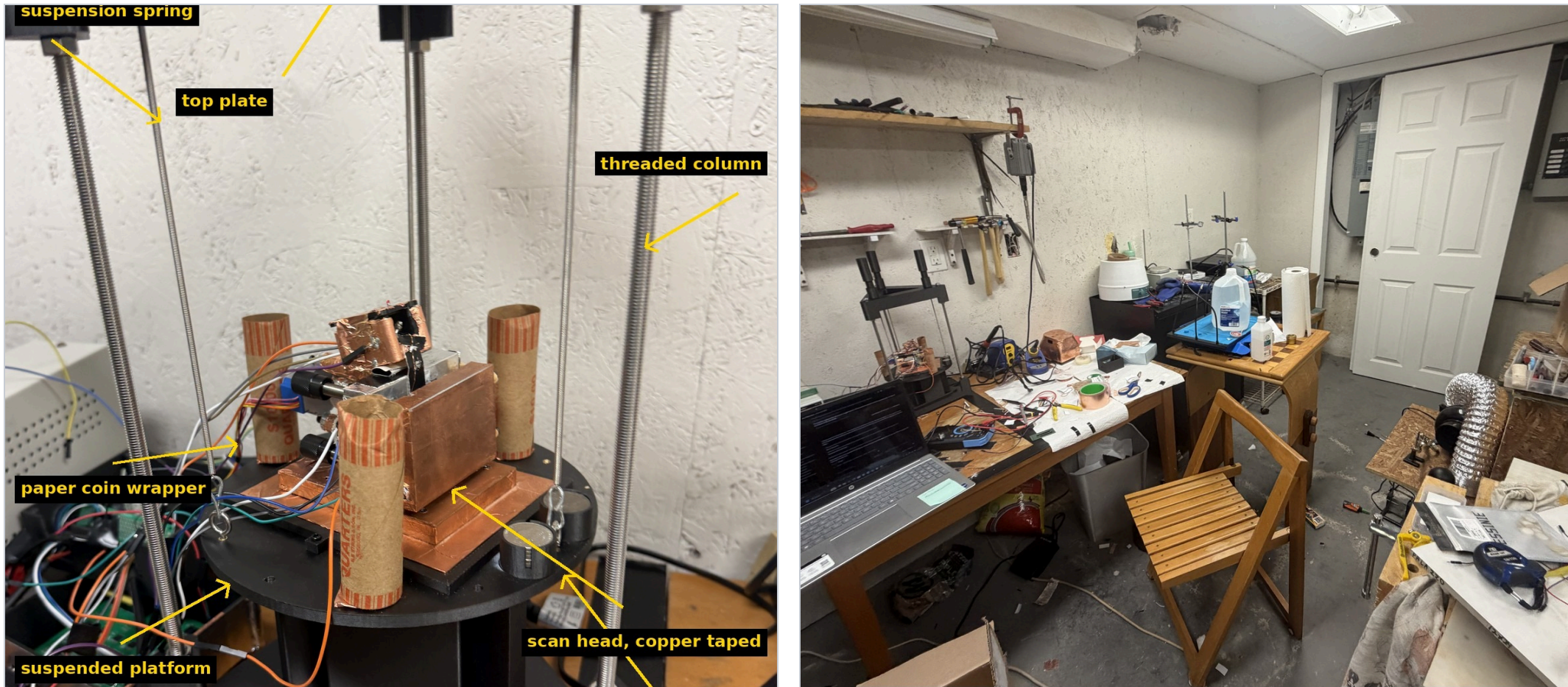


Nuh Shaheer, left. Jacob Katz, right. The morning of the last bench session, with the instrument still assembled behind them. Identification and permission to publish **SAID** by Jacob, 2026-09-19 (Nuh, second-hand via Jacob, 2026-09-20). IMG_8633.

Methods: The Instrument

We built the whole thing

There was no kit. We ordered the boards, printed the frame, adapted the firmware, and brought up every stage on its own before trusting the next one.



Left: the instrument on the morning it came apart to be moved. Printed frame; springs down to the suspended platform; copper-taped scan head on a copper ground plane. **Right:** where it happened — a wooden bench on a concrete basement floor, a few meters from an electrical panel. **READ.** 2026-09-19.

A Teensy 4.1 drives four 16-bit converters over a 1 MHz bus and a 26-way ribbon — X, Y and sample bias at ± 3 V, Z at ± 10 V — moving a piezoelectric disc that carries a cut tungsten tip. The tip current goes into a transimpedance amplifier with a 100 M Ω feedback resistor, in a hand-folded copper box, and is read back by a 16-bit LTC2326-16. Coarse approach is a 28BYJ-48 stepper on a lever. **Mechanics, controller and firmware follow Mech Panda's *red-panda-stm* [1]; the scan head and preamplifier follow Dan Berard's home-built STM [2].** The electronics bring-up, the software, the calibration and the controls are ours.

Key Results

0.13 σ

The chain agrees with theory

Measured gain against what Ohm's law required, worked out before the run. 53 readings. **MEASURED** 2026-09-16

4 pA

Amplifier's own input current

Against the ~ 1 nA it has to find — 250 times smaller. **MEASURED** 2026-09-15

16–59 M Ω

A real junction, made and swept

Resistance falls as the voltage rises — not a short, not an open. **MEASURED** 2026-09-17

9–19

Counts of noise with a junction live

Our electronics are not what limits us, and we proved it with numbers. **MEASURED** 2026-09-19

60,928

Readings inside the tunneling window

Taken across 109 separate approaches, with bias on the whole way. **MEASURED** 2026-09-19

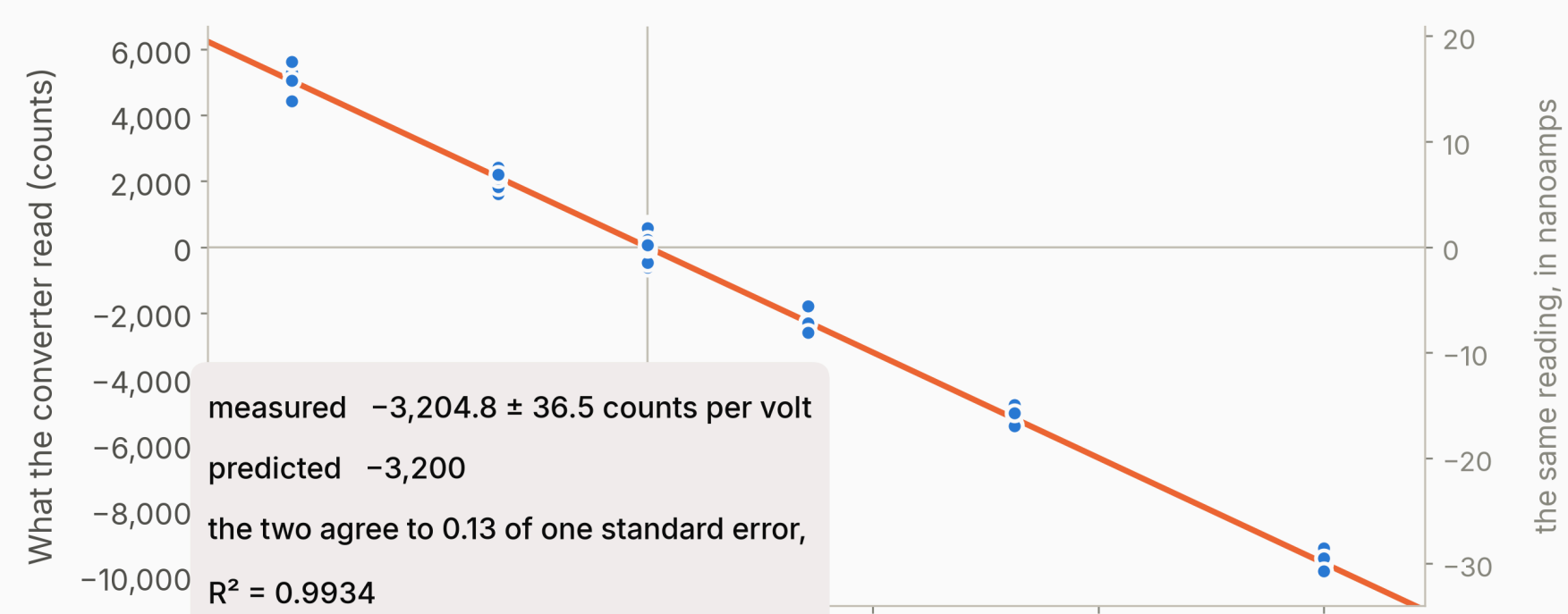
Results: Calibration of the Measurement Chain

The measurement chain agrees with theory

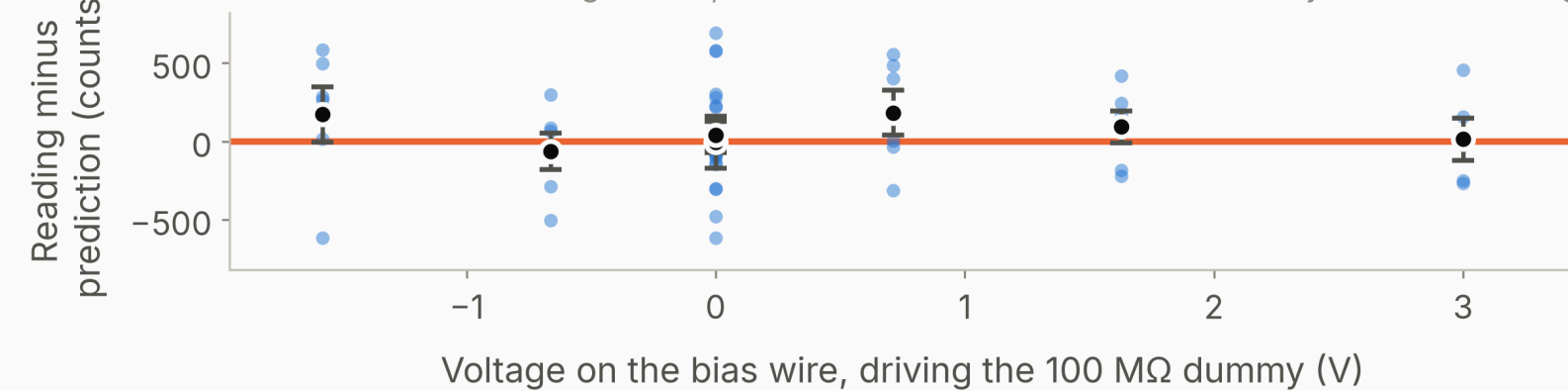
We took the tip and sample out and clipped a **known 100 M Ω resistor** where the junction would be. That makes the current something Ohm's law fixes **before the run** — so the instrument is checked against a number it had no part in choosing.

The whole measurement chain works end to end — and agrees with theory to 0.15%

A known 100 M Ω resistor stood in for the tunneling junction. **The 53 readings landed on the line Ohm's law requires**, which was worked out before the bench run and is not fitted to anything.



Dark dots are each setting's mean, with its standard error. Pale dots are every individual reading.



Source: sessions/2026-09-16-bench.md section 3.4 (no CSV exists for that date). Measured 2026-09-16, 19:23–19:25 UTC, cover off, no tip fitted.

WHAT THIS DOES NOT SHOW: the slope is the ratio of the feedback resistor to the dummy, both nominal 100 M Ω , neither tolerance recorded — so this confirms the scale to within those tolerances, not independently of them. The 53 readings are repeats at 8 settings, not 53 independent observations. The 320-count scatter is not a noise figure (cover off, clips on the input).

Measured $-3,204.8 \pm 36.5$ counts per volt against $-3,200$ predicted — agreement to 0.13 of one standard error, $R^2 = 0.9934$ over 53 readings. Twelve stages had to be right at once for those points to land on that line. Source: sessions/2026-09-16-bench.md §3.4. **What it does not show:** the slope is the ratio of two nominally 100 M Ω resistors whose tolerances were never recorded; and the 53 readings are repeats at 8 settings, not 53 independent observations.

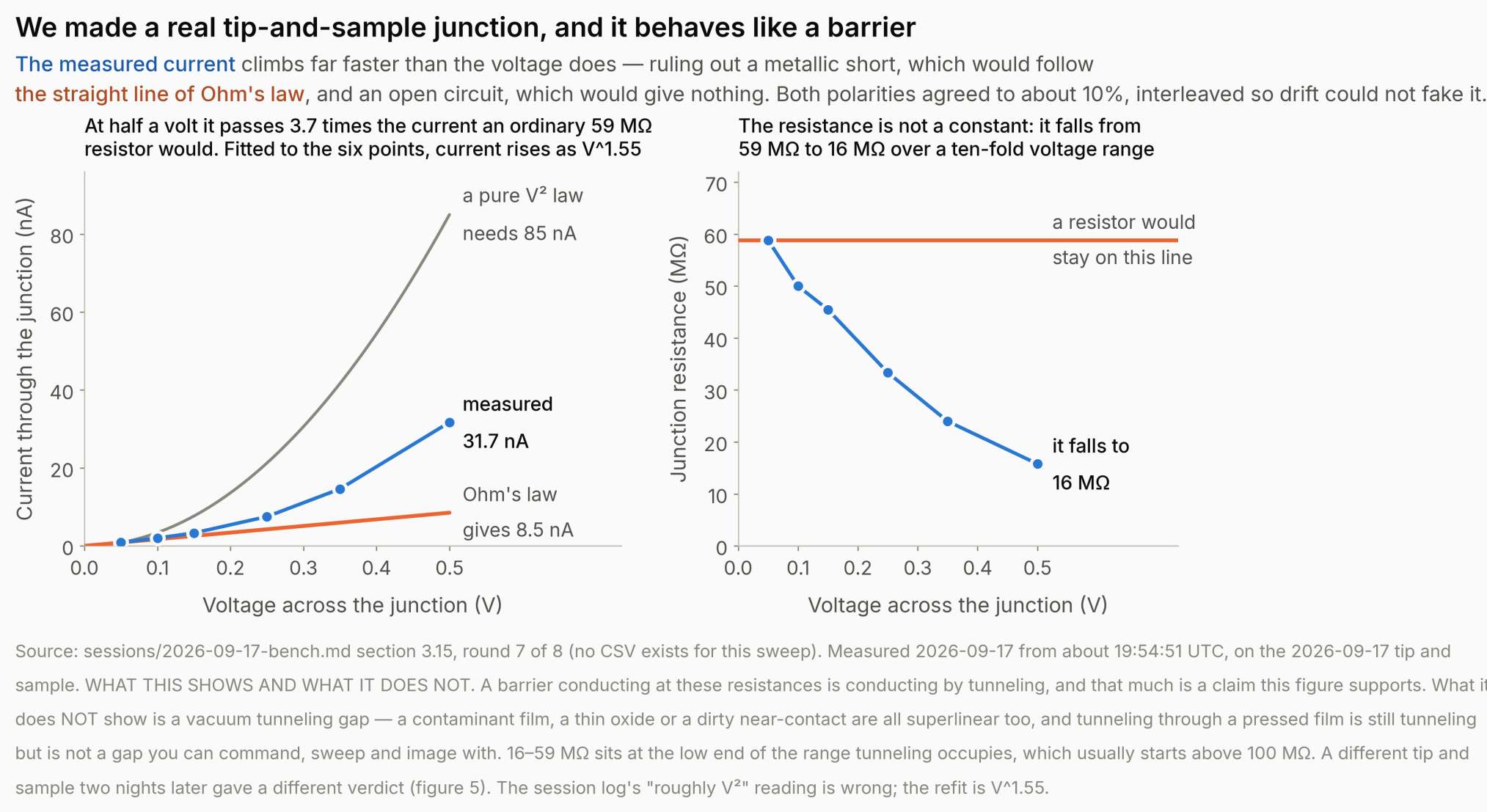
This is the number everything else rests on

It fixes the conversion — 320.5 counts per nanoamp — and it settles the sign, which matters because a current that drove the reading the unexpected way would send an automatic approach straight into the sample.

Results: A Real Junction

A real junction, with a barrier's signature

Getting a current is easy — push the tip in and there is plenty. The hard part is showing you have a **barrier**: two conductors with something in between that electrons have to cross.



Source: sessions/2026-09-17-bench.md section 3.15, round 7 of 8 (no CSV exists for this sweep). Measured 2026-09-17 from about 19:54:51 UTC, on the 2026-09-17 tip and sample. WHAT THIS SHOWS AND WHAT IT DOES NOT. A barrier conducting at these resistances is conducting by tunneling, and that much is a claim this figure supports. What it does NOT show is a vacuum tunneling gap — a contaminant film, a thin oxide or a dirty near-contact are all superlinear too, and tunneling through a pressed film is still tunneling but is not a gap you can command, sweep and image with. 16–59 M Ω sits at the low end of the range tunneling occupies, which usually starts above 100 M Ω . A different tip and sample two nights later gave a different verdict (figure 5). The session log's "roughly $V^{1.55}$ " reading is wrong; the left is $V^{1.55}$.

Current rose from 0.85 nA at 0.05 V to **31.7 nA at 0.5 V** — **3.7 times faster than Ohm's law**, fitting $V^{1.55}$, with the junction resistance falling from 59 to 16 M Ω . Eight rounds with the polarities interleaved, so drift could not manufacture a slope; symmetric both ways to about 10%; confirmed by a bias-flip test on two separate nights. **That rules out a metallic short, which would be ohmic, and an open circuit, which would give nothing.** Source: sessions/2026-09-17-bench.md §3.15.

And the electronics are not the limit

With a junction live the noise floor is **9–19 counts**; leakage through the tip lead is under 0.06 nA across ± 2 V, more than 8 G Ω ; and someone stamping on the floor two meters away moves one reading's spread from 131 to 132 pA. **The building is not getting in electrically.**

Limitations

What we do not claim

No image. Fifty-one recordings, none of them a demonstrable image of the sample — and our own controls are how we know. **No controlled vacuum gap**, which is the regime you need in order to scan: we measured tunneling through a barrier junction, and holding a gap open is a different thing that we have not shown. The 2026-09-19 junction was a pressed contact, measured. **No atomic resolution and no scale bar anywhere** — every axis is in converter counts, because this instrument has never established a distance scale of its own. No number is read off a photograph: captions are marked SAID or READ.

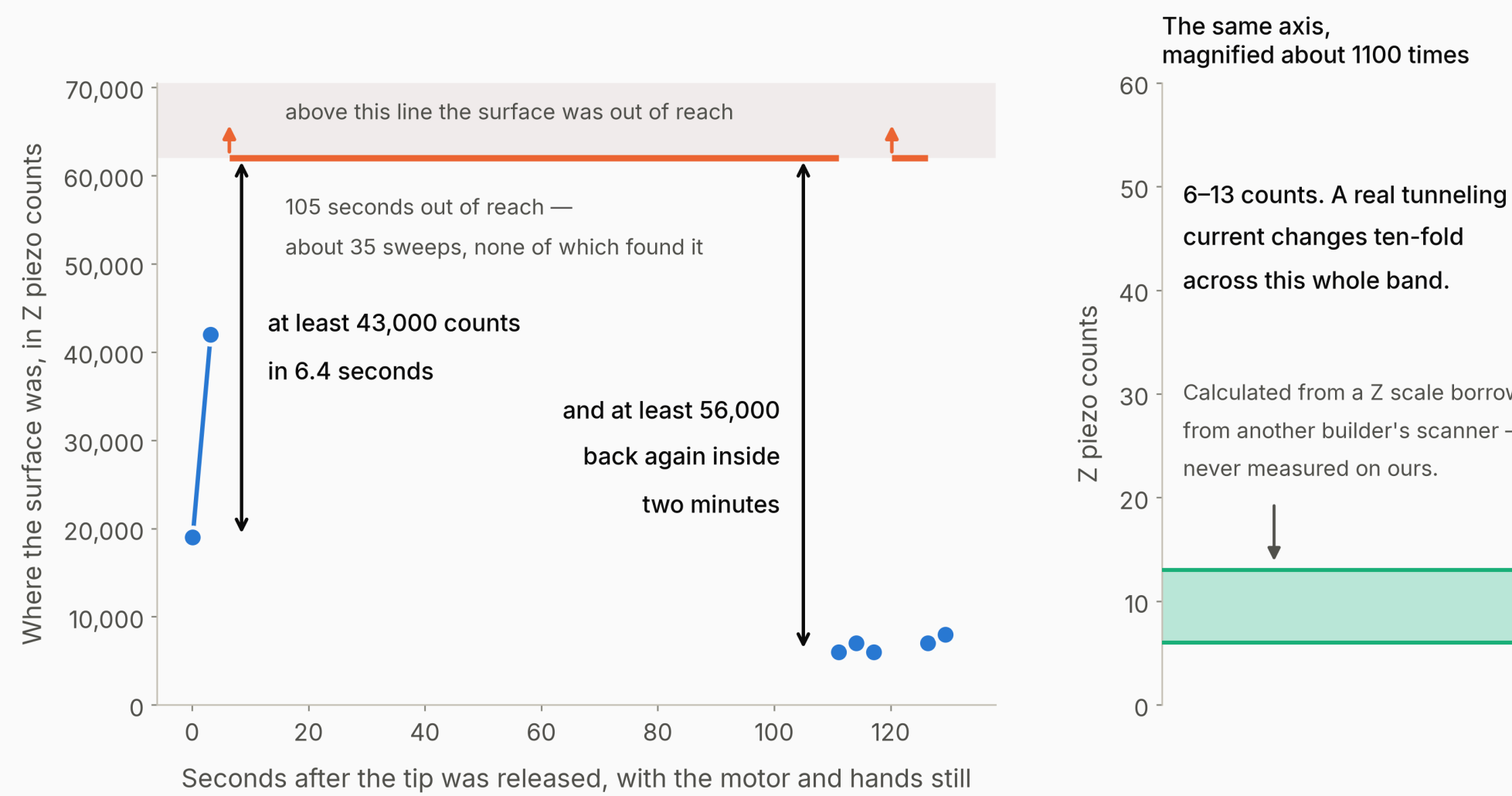
Results: Stability of the Gap

We found the limit, and measured it

To take an image the gap has to hold still while the tip sweeps. We stopped the motor, took our hands off the instrument, and watched where the surface went.

We found what is stopping us, and measured it

With the motor off and nobody touching the instrument, **the surface the tip was hunting for** moved by most of the piezo's whole range within seconds — and **went out of reach altogether** for nearly two minutes. Knowing this, precisely, is the result.



Source: sessions/data/2026-09-19-morning/release_watch_run1.log, parsed line by line by this script. Measured 2026-09-19, 12:54:10–12:56:19 UTC; sessions/2026-09-19-morning.md section 3.9. The orange out-of-reach bars are not printed points: the watcher logs only when the state changes (release_and_watch.py) and swept every 3 s throughout.

THE FAIR QUESTION: the first excursion begins 0.1 s after a 60-step motor retract ended, so relaxation after that move cannot be excluded for that one event — but the surface reverses direction and overshoots its own starting point, which relaxation does not do, and the 56,000-count figure spans 6.4 s to 111 s.

WHAT THIS DOES NOT SHOW: what is moving, or why — still UNKNOWN; nine events are too sparse to say whether it is periodic. Onsets are multiples of 1,000 counts because the sweep step was stuck there. The 6–13 band is calculated from an inherited Z scale, and no distance scale appears here because this instrument has never established one from its own hardware. The blue dots are contact points, not an image — this project has never produced an image.

The surface moved by more than 43,000 Z counts within 6.4 seconds, and by more than 56,000 within two minutes — most of the piezo's whole range — and went out of reach altogether for 105 seconds. Source: sessions/data/2026-09-19-morning/release_watch_run1.log. **The fair question:** the fast excursion begins 0.1 s after a motor retract ended, so relaxation cannot be excluded for that one event — but the motion reverses and overshoots its own starting point, which relaxation does not do, and the two-minute figure is measured long after any transient.

Why that ends the imaging question, in one comparison

Over the five seconds one image takes, the gap drifts **396 to 1,722 counts**. The "structure" in every candidate image we have is **29 to 458 counts**. **The thing being measured is smaller than the way the gap moves while it is being measured.** Knowing that, precisely, is the result.

Results: Controls and Self-Audit

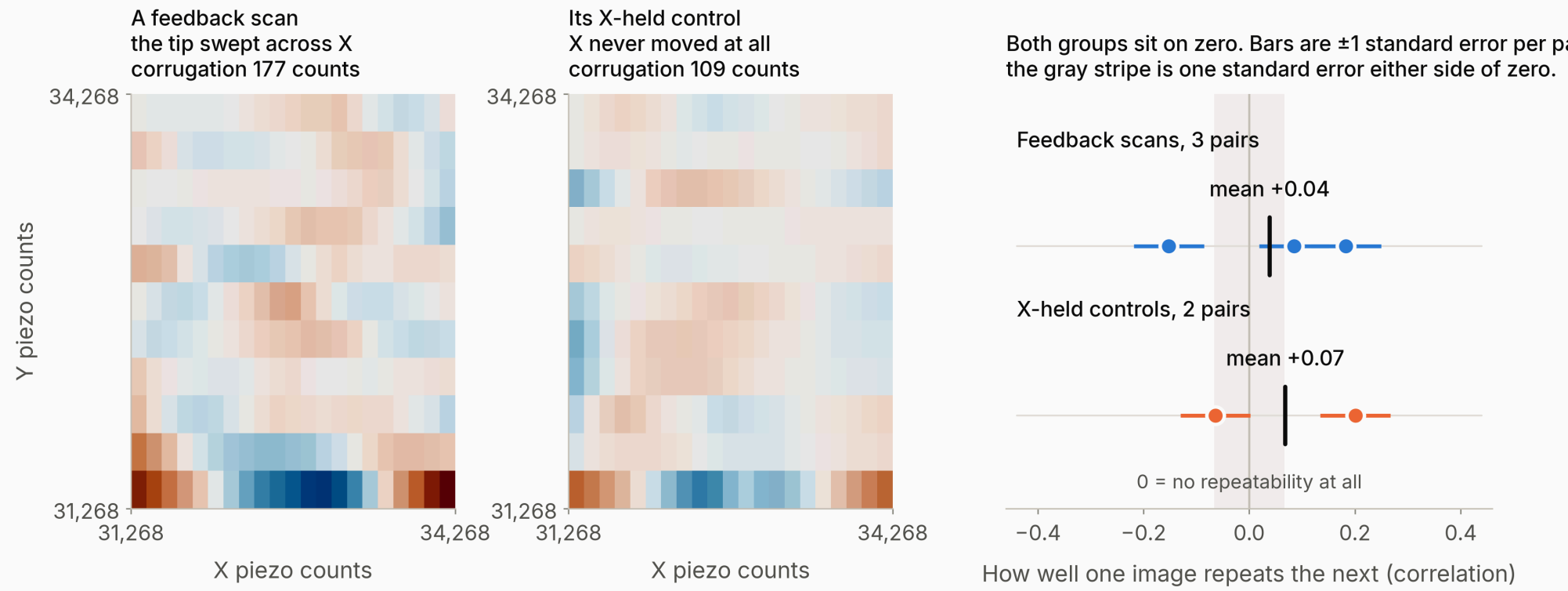
We built controls designed to kill our own results

Every scan was run again with the sideways sweep switched off. An **X-held control** has identical timing and identical loop behavior, but the tip never travels across the surface — so it **cannot contain surface structure at all, by construction**.

The control that stopped us fooling ourselves

After every scan we ran it again with the lateral sweep switched off, so **the control** could not contain surface structure at all.

The real scans repeat no better than those controls do. That one test is what settles the imaging question — and running it was the right call.



NEITHER PANEL IS AN IMAGE OF A SURFACE. Each is the Z code the feedback loop asked for at that point, after the line's own tilt is removed (counts). Zero is that line's own average.

Source: sessions/data/2026-09-19-bench/cas9_scan0-3.csv and cas9_xheld0-3.csv, measured 2026-09-19 from about 03:48 UTC; sessions/2026-09-19-bench.md section 3.14. Every number here is recomputed from those files by this script.

CORRECTION CARRIED: the previously published control mean of $+0.37$ came from cas9_xheld2.csv, a ONE-LINE file whose run aborted, silently truncated to 21 points against another image's first line (LEAD_VERIFICATION.md v4). Only full 231-point images are used here, and this script refuses to correlate two images of different length. The honest comparison is $+0.04$ against $+0.07$. NEITHER PANEL IS AN IMAGE OF A SURFACE. They are the Z code the feedback loop asked for at each point, with each line's tilt removed; the color map is Cramer's vlik, centered on zero, and nothing is interpolated or smoothed. Axes are in DAC counts because this instrument has never established a distance scale from its own hardware.

WHAT THIS DOES NOT SHOW: with 3 and 2 pairs, neither mean is distinguishable from zero or from the other — the comparison was never powered to support "better", in either direction.

Like for like over full images, our scans repeat at **± 0.04** and their controls at **± 0.07** : indistinguishable, and neither differs from zero. That single test is what settles the imaging question, and running it was the right call. Source: sessions/data/2026-09-19-bench/cas9_*.csv.

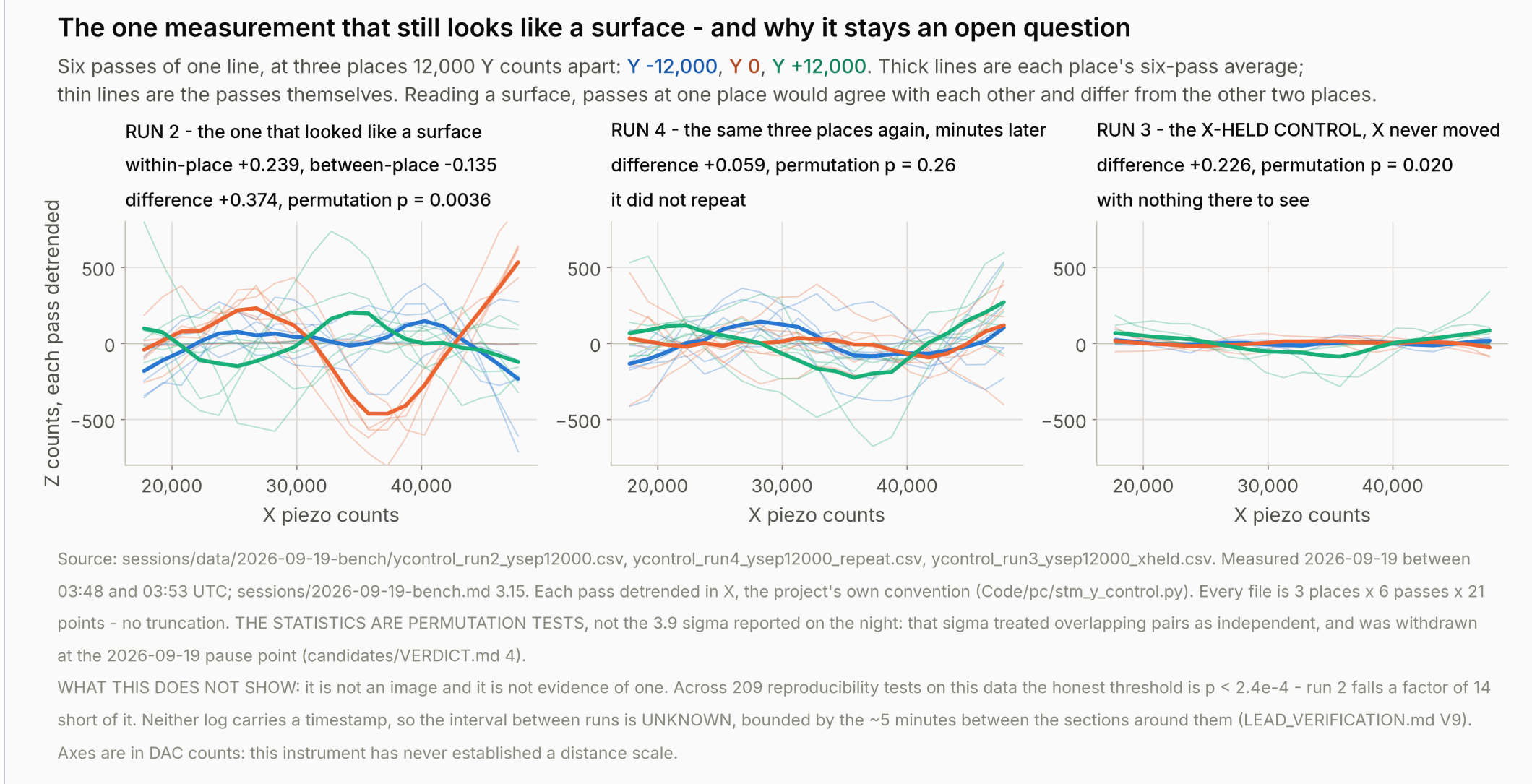
We audited ourselves, and published what we found

At this pause point we re-derived our own published numbers from the raw files. **Two statistics we had used to dismiss our own scans turned out to be computed across aborted files**, and a reproducible feature we found most convincing turned out to be the feedback loop recovering from a jump, not the sample. **Every conclusion survived. Two of the arguments did not** — and both withdrawals are in the public record.

Discussion

We detected tunneling — and we know what stopped us holding it

Our best junction, two nights before the pressed contact, changed ten-fold about **seven times faster**: roughly 250 Z counts per decade.



Source: sessions/data/2026-09-19-bench/ycontrol_run2_ypiezo/12000.csv, ycontrol_run4_ypiezo/12000_repeat.csv, ycontrol_run3_ypiezo/12000_xheld.csv. Measured 2026-09-19 between 03:48 and 03:53 UTC; sessions/2026-09-19-bench.md section 3.15. Each pass detrended in X, the project's own convention (Code/pclstm_ycontrol.py). Every file is 3 places $\times$ 6 passes $\times$ 21 points - no truncation. THE STATISTICS ARE PERMUTATION TESTS, not the 3.9 sigma reported on the right: that sigma treated overlapping pairs as independent, and was withdrawn at the 2026-09-19 pause point (cas9_xheld0-3.csv v4). WHAT THIS DOES NOT SHOW: it is not an image and it is not evidence of one. Across 209 reproducibility tests on this data the honest threshold is $p < 2.4 \times 10^{-4}$. - run 2 falls a factor of 14 short of it. Neither log carries a timestamp, so the interval between runs is UNKNOWN, bounded by the ~ 5 minutes between the sections around them (LEAD_VERIFICATION.md v9). Axes are in DAC counts; this instrument has never established a distance scale.

The one measurement that still looks like a surface. Six passes at each of three places: passes at the same place agreed **± 0.239** while different places disagreed **-0.135** . The repeat, minutes later, gave **± 0.056** — and an X-held control, which cannot contain a surface, scored $p = 0.020$ on the same statistic. Against 209 reproducibility tests the honest threshold is $p < 2.4 \times 10^{-4}$. **Not an image, and not presented as one.**

We detected tunneling. We could not hold the range long enough to scan

A tip cannot go from not touching to touching without passing through the separations where the only thing carrying electrons is tunneling — and we had bias applied and the amplifier recording the whole way. **60,928 readings inside that range across 109 approaches.** The current is superlinear and symmetric and the junction never metallically shorted: a **barrier, conducting by tunneling. What we have not done is hold that gap open and steady enough to scan.** That turns on d, the tip's distance from the pivot line, which sets the lever ratio: it wants to be near 26 μm , and we have it bounded at under 0.5 mm. **A microscope closes it in minutes.**

The Second Instrument: The System That Runs the Science

Neither of us writes code. So the other thing we built was the framework that does — and, more to the point, the framework that catches it when it is wrong.

Two people, two computers, days apart. **The failure that kills a project like this is not a hard experiment. It is a number corrected in one document and left standing in the eleven others that quote it.**

That is not hypothetical here. On 2026-09-07 two constants changed, both were already written into more than a dozen documents, and correcting every copy anyone could think of **still left stale values in six files**.

So every number in this project now lives in exactly one place, carries a tag saying how it is known — **MEASURED, CALC, DS, MESH, ORDER, SAID, READ** — and is checked by a program, not by memory.

What the framework is made of

The protocol what the model must read first	575 lines
The register one canonical value per fact	157 rows
The checker seven automated tests, every session	598 lines
Session logs append-only, never rewritten	29
Reference documents	18
Bench tools Python, talking to the instrument	16

What proves it works is the retractions

- A statistical result computed across truncated files — withdrawn, and the same defect then swept across every other comparison in the project.
- Our most convincing reproducible feature — shown to be the feedback loop recovering from a flyback, predicted to the exact count. Withdrawn.
- A detector's full-scale voltage recorded backwards in a live reference document. Caught by the checker.
- Eleven wrong numbers in one commit, found by a verification pass and fixed before the second push.

An AI that writes confident prose is not hard to get. One that produces a retraction is, and that is the part we built. We did not train a model — the model is off the shelf. The discipline around it is ours.

Future Work

Where it goes next

- Measure d properly — a microscope or comparator, no power, and it settles our best junction retroactively.
- Prove the gap holds still before scanning anything — a box over the instrument, a 15-minute recording, a gate of 200 counts over sixty seconds.
- A stiffer sample than gold leaf on paper, and a sharper tip.
- Three free software fixes that remove the artifact behind our most convincing false positive: drop the first three pixels of each pass, sweep X both ways between passes, never compare two images of different length.

References

- Mech Panda. *red-panda-stm* — mechanics, controller board and firmware. github.com/MechRedPanda/red-panda-stm
- D. Berard. *Home-built STM* — scan head, tip preparation and transimpedance preamplifier. dberard.com/home-built-stm/ © 2017 Daniel Berard; the preamplifier Eagle files are MIT licensed.
- J. Katz and N. Shaheer. *We Are STMING* — every session log, raw data file and analysis script behind the figures on this poster. github.com/jacobkatz/We-Are-STMING

Where a figure caption names a session log or a raw data file, that file is in [3].

Acknowledgments

Mech Panda (*red-panda-stm*: mechanics, controller, firmware) and **Dan Berard** (scan head, preamplifier) published their instruments for free. Without them there would have been nothing to bring up.

Every session log, every raw file, every analysis script and every withdrawn claim: github.com/jacobkatz/We-Are-STMING